

SaaS Financial Performance, Unit Economics & Churn Risk Analysis

SaaS Financial Performance, Unit Economics & Revenue Leakage Analysis

In the subscription economy, topline revenue metrics like ARR can obscure critical underlying churn risks. This end-to-end financial data analysis evaluates portfolio durability, contract concentration risk, and churn-driven financial leakage for a SaaS business managing **\$5.47M in Annual Recurring Revenue (ARR)** across 7,043 active accounts. By combining an optimized Python/Pandas data pipeline with an in-memory ANSI SQL aggregation engine (**DuckDB**), this project bridges technical data engineering and strategic Financial Planning & Analysis (FP&A). The analysis pinpoints core operational bottlenecks—isolating **\$1.67M in annual revenue leakage**—and delivers C-suite executive recommendations to stabilize cash flows and improve unit economics.

1.Code:

```
import pandas as pd
import numpy as np

url="https://raw.githubusercontent.com/IBM/telco-customer-churn-on-icp4d/master/data/Telco-Customer-Churn.csv"
df = pd.read_csv(url)

df['TotalCharges'] =
pd.to_numeric(df['TotalCharges'].str.strip,errors='coerce')
df['TotalCharges'] = df['TotalCharges'].fillna(0)

df['MRR'] = df['MonthlyCharges']
df['ARR'] = df['MonthlyCharges']*12

print(f"Total Portfolio ARR: ${df['ARR'].sum():,.2f}")
print(f"Total Portfolio MRR: ${df['MRR'].sum():,.2f}")
```

Output:

```
Total Portfolio ARR: $5,473,399.20
Total Portfolio MRR: $456,116.60
```

Keyfinding:

Executive Financial Summary (Step 1):

- **Portfolio ARR Scale:** The company currently manages an Annual Recurring Revenue (ARR) base of **\$5.47M** (\$5,473,399.20), derived from a Monthly Recurring Revenue (MRR) base of **\$456.12K** (\$456,116.60) across 7,043 active subscriptions.
- **Run-Rate Baseline:** The average monthly billing engine generates approximately **\$456K/month**, providing a stable baseline for recurring revenue forecasting and OpEx planning.
- **Data Integrity Note:** All raw transaction values were sanitized (TotalCharges missing values imputed to \$0 for newly onboarded tenure-0 accounts), ensuring 100% data integrity for revenue metrics.

2.Code:

```
# 2. Step: Contract Grouping
contract_analysis = df.groupby('Contract').agg(
    Customer_Count=('customerID', 'count'),
    Total_ARR=('ARR', 'sum'),
    Avg_MRR=('MRR', 'mean')
).reset_index()

total_company_arr = df['ARR'].sum()
contract_analysis['ARR_Share_%'] = (contract_analysis['Total_ARR'] /
total_company_arr) * 100

contract_analysis['Total_ARR'] = contract_analysis['Total_ARR'].round(2)
contract_analysis['Avg_MRR'] = contract_analysis['Avg_MRR'].round(2)
contract_analysis['ARR_Share_%'] = contract_analysis['ARR_Share_%'].round(2)

# --- OUTOUT: MUST USE PRINT STATEMENT ---
print("\n" + "="*50)
print("=== Step 2: Revenue Concentration by Contract Type ===")
print("="*50)
print(contract_analysis)
print("="*50 + "\n")
```

output:

```
=====
=== Step 2: Revenue Concentration by Contract Type ===
=====
```

	Contract	Customer_Count	Total_ARR	Avg_MRR	ARR_Share_%
0	Month-to-month	3875	3087529.8	66.40	56.41
1	One year	1473	1149799.2	65.05	21.01
2	Two year	1695	1236070.2	60.77	22.58

Keyfinding:

Executive Financial Summary (Step 2 - Concentration & Risk Analysis):

- **High Churn Exposure:** The portfolio exhibits a high revenue concentration risk, with **56.41% (\$3.09M ARR)** tied to short-term **Month-to-month** contracts. This signifies that over half of the company's recurring revenue stream is vulnerable to immediate cancellations.
- **Contract Stability Gap:** Long-term committed revenue (1-Year and 2-Year contracts) accounts for only **43.59% (\$2.39M ARR)** of total annual revenue, highlighting a critical need for incentivizing long-term contract lock-ins.
- **Pricing Dynamics:** Month-to-month subscribers generate the highest average MRR (\$66.40), whereas 2-Year contract customers generate \$60.77/month, reflecting a ~8.5% implicit discount offered in exchange for revenue predictability and lower churn risk.

3.Code:

```
# === STEP 3: Churn & Financial Leakage Analysis ===

# 1. Aggregation Engine
churn_analysis = df.groupby('Churn').agg(
    Customer_Count=('customerID', 'count'),
    Total_ARR=('ARR', 'sum'),
    Avg_MRR=('MRR', 'mean')
).reset_index()

# 2. Vectorized Percentage Computation
total_company_arr = df['ARR'].sum()
churn_analysis['ARR_Share_%'] = (churn_analysis['Total_ARR'] /
total_company_arr) * 100

# 3. Floating Point Precision & Formatting
churn_analysis['Total_ARR'] = churn_analysis['Total_ARR'].round(2)
churn_analysis['Avg_MRR'] = churn_analysis['Avg_MRR'].round(2)
churn_analysis['ARR_Share_%'] = churn_analysis['ARR_Share_%'].round(2)

print(churn_analysis)
```

output:

Churn	Customer_Count	Total_ARR	Avg_MRR	ARR_Share_%	
0	No	5174	3803829.0	61.27	69.5
1	Yes	1869	1669570.2	74.44	30.5

Keyfinding:

Executive Financial Summary (Step 3 - Churn & Financial Leakage):

- **Severe ARR Leakage:** The business suffers a massive financial leakage of **\$1.67M ARR (30.5% of total portfolio revenue)** attributed to 1,869 churned accounts.
- **Premium Segment Erosion:** Churned subscribers exhibit a significantly higher average monthly bill (**\$74.44 Avg MRR**) compared to retained subscribers (**\$61.27 Avg MRR**). This indicates an accelerated loss of high-tier, higher-value customer segments.
- **Revenue Churn vs. Customer Churn Gap:** While customer churn stands at 26.54%, revenue churn reaches 30.50%. This 3.96% discrepancy confirms that churn is disproportionately impacting top-tier revenue-generating accounts, signaling potential pricing fatigue or value-realization issues in premium tiers.

4.Code:

```
# === STEP 4: Customer Cohort & Tenure Analysis ===

# 1. Define Tenure Bins and Labels using pd.cut
bins = [-1, 12, 24, 48, 72]
labels = ['0-1 Year (0-12m)', '1-2 Years (13-24m)', '2-4 Years (25-48m)', '4-6 Years (49-72m)']

df['Tenure_Cohort'] = pd.cut(df['tenure'], bins=bins, labels=labels)

# 2. Aggregation Engine: Cohort-level Financial Breakdown
cohort_analysis = df.groupby('Tenure_Cohort', observed=False).agg(
    Total_Customers=('customerID', 'count'),
    Churned_Customers=('Churn', lambda x: (x == 'Yes').sum()),
    Total_ARR=('ARR', 'sum'),
    Avg_MRR=('MRR', 'mean')
).reset_index()

# 3. Vectorized Rate Computations
cohort_analysis['Churn_Rate_%'] = (cohort_analysis['Churned_Customers'] /
cohort_analysis['Total_Customers']) * 100
cohort_analysis['ARR_Share_%'] = (cohort_analysis['Total_ARR'] /
df['ARR'].sum()) * 100

# 4. Precision Formatting
cohort_analysis['Churn_Rate_%'] = cohort_analysis['Churn_Rate_%'].round(2)
cohort_analysis['ARR_Share_%'] = cohort_analysis['ARR_Share_%'].round(2)
cohort_analysis['Total_ARR'] = cohort_analysis['Total_ARR'].round(2)
cohort_analysis['Avg_MRR'] = cohort_analysis['Avg_MRR'].round(2)
```

```
print("\n" + "="*70)
print("=== Step 4: Customer Cohort & Tenure Analysis ===")
print("="*70)
print(cohort_analysis)
print("="*70 + "\n")
```

output:

```
=== Step 4: Customer Cohort & Tenure Analysis ===
=====
```

Tenure_Cohort		Total_Customers	Churned_Customers	Total_ARR	Avg_MRR
Churn_Rate_%	ARR_Share_%				
0	0-1 Year (0-12m)	2186	1037	1471557.0	56.10
47.44	26.89				
1	1-2 Years (13-24m)	1024	294	753958.2	61.36
28.71	13.77				
2	2-4 Years (25-48m)	1594	325	1261119.6	65.93
20.39	23.04				
3	4-6 Years (49-72m)	2239	213	1986764.4	73.95
9.51	36.30				

Keyfinding:

Executive Financial Summary (Step 4 - Tenure Cohort & Lifecycle Dynamics):

- **Onboarding Churn Bottleneck:** The **0-1 Year (0-12m)** cohort exhibits an alarming **47.44% churn rate** (1,037 accounts lost out of 2,186). This indicates severe friction during the onboarding lifecycle, causing high initial Customer Acquisition Cost (CAC) decay before margin recovery.
- **LTV Expansion in Mature Cohorts:** Retention improves drastically with account age. The **4-6 Years (49-72m)** mature cohort demonstrates the lowest churn rate at **9.51%** while driving the highest ARPU (**\$73.95 Avg MRR**), accounting for **36.30% (\$1.99M ARR)** of the total revenue base.
- **Strategic Takeaway:** Capital allocation should prioritize early-stage customer success programs and onboarding support rather than pure top-of-funnel acquisition, as securing accounts past the 12-month mark drops churn probability by over 40 percentage points.

5.Code:

```
import duckdb

# === STEP 5: SQL Automation Engine (DuckDB Integration) ===

# 1. SQL Query for Executive Metric Summaries
sql_query = """
SELECT
    Contract,
    COUNT(customerID) AS Total_Customers,
    SUM(CASE WHEN Churn = 'Yes' THEN 1 ELSE 0 END) AS Churn_Count,
    ROUND(SUM(CASE WHEN Churn = 'Yes' THEN 1 ELSE 0 END) * 100.0 /
COUNT(customerID), 2) AS Churn_Rate_Pct,
    ROUND(SUM(ARR), 2) AS Contract_ARR,
    ROUND(SUM(CASE WHEN Churn = 'Yes' THEN ARR ELSE 0 END), 2) AS
Financial_Leakage_ARR,
    ROUND(AVG(MRR), 2) AS Avg_MRR
FROM df
GROUP BY Contract
ORDER BY Financial_Leakage_ARR DESC;
"""

# 2. Execute SQL query directly against the Pandas DataFrame 'df'
sql_executive_summary = duckdb.query(sql_query).df()

print("\n" + "="*80)
print("=== Step 5: Automated SQL Executive Summary (DuckDB Engine) ===")
print("="*80)
print(sql_executive_summary)
print("="*80 + "\n")
```

output:

```
=====
==
=== Step 5: Automated SQL Executive Summary (DuckDB Engine) ===
=====
==
      Contract  Total_Customers  Churn_Count  Churn_Rate_Pct  Contract_ARR
Financial_Leakage_ARR  Avg_MRR
0  Month-to-month           3875         1655.0           42.71      3087529.8
1450165.2      66.40
1      One year           1473          166.0           11.27      1149799.2
169421.4      65.05
2      Two year           1695           48.0            2.83      1236070.2
49983.6      60.77
```

Keyfinding:

Executive Financial Summary (Step 5 - Contract Leakage Isolation & Root Cause Analysis):

- **Concentrated Revenue Vulnerability:** DuckDB SQL aggregation isolates **\$1.45M (86.86%)** of total portfolio ARR leakage (\$1.67M) directly to the **Month-to-month** contract segment, which suffers a catastrophic **42.71% segment churn rate**.
- **Contract Lock-in Efficacy:** Multi-year contract structures act as a powerful financial shield. **Two-year contracts** restrict annual revenue loss to just **\$49.98K (2.83% churn rate)**, despite representing **\$1.24M** in total portfolio exposure.
- **Strategic Shift:** Management must prioritize contract conversion initiatives (e.g., offering targeted 5-10% annual migration discounts). Transitioning just 20% of Month-to-month subscribers into 1-Year or 2-Year plans would immediately safeguard **~\$290K+ in annual recurring revenue**.